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# Privacy Analysis of Augmented Reality Glasses

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### Weekly Report 5

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**Michelle Jessica Bösigler**  
michellej.boesiger@unibas.ch

Department of Mathematics and Computer Science  
University of Basel

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#### What has been done (Previous Week)

- Had a meeting with Nima on Monday to discuss the issues with Adafruit and the Micro:bits. The micro:bit was able to pick up something whilst on the second floor of the Uni building but otherwise was the same as before.
- Mainly - been trying to get both the Micro:bit and Adafruit to sniff the Bluetooth traffic. Still unsuccessful sadly. So the following is just a quick rundown on what I tried throughout the week (at least the things I can remember right now as I tried a lot of random things that lead nowhere).
- I went to look into the hardware components to try and see if it could be a compatibility issue.
  - At first I thought it may be because the glasses are using both Classic Bluetooth and BLE and just that it is not using BLE enough to be shown. That was my initial assumption as to why it did not work, but my issue with that is that micro:bits is not able to capture anything which should not be (the Adafruit is able to sniff something, just not the glasses).
- I tried directly capturing on Wireshark and use either of the two as the capture interface.
  - I was not able to get either of the device to be recognized and show up as an interface (confirmed that with TShark). According to the internet it could be because they are not compatible or because the setup is missing, so I tried doing the setup to check but it still did not work.
  - Also the Bluetooth monitor on my laptop is not capable of BLE sniffing as no traffic was being captured
- I thought it could be because the MAC address is different on different devices (although chances are slim)
  - I was able to connect the glasses to my laptop temporarily and wrote my own code for my laptop to scan for BLE and the glasses was displayed with the same address, so that's cleared. Sad that I can't have the Meta View app on the laptop for the functionalities.
- I thought it could really just be the set up
  - Adafruit - I tried to write my own code to scan for nearby BLE. From the git repository, [https://github.com/adafruit/Adafruit\\_nRF52\\_Arduino.git](https://github.com/adafruit/Adafruit_nRF52_Arduino.git), I set up the Arduino IDE on my laptop but it just created more issues than it solved (adafruit-nrfutil).

- Micro:bit - Btlejack (according to the internet) is the main one used as there aren't others that would be able to offer what btlejack offer, so I just tried reflashing the microbits (also in different ways) to try it out but still have the same problem. I tried sniffing the traffic using each micro:bit individually and then only two at a time, but as expected that did not do anything.
- I remembered a brief mention about rooting the phone and so I looked into that.
  - From a forum XDA, <https://xdaforums.com/t/guide-all-about-unlocking-bootloader-twrp-and-rooting-redmi-note-8-8t.4031831/>, they mention that in trying to unlock the Bootloader if I want to 'unlock' in one of the steps the time usually needed to wait is 168 h, which since it is in Fastboot mode, I will not be able to do anything else. I wanted to see if it would work if I don't do that but then another step would erase all data on the phone, which I do not know can I do that. I did not read that in time, btw, that that would happen so I am glad that the command to do it failed.
  - I also tried to directly capture the Bluetooth traffic directly on the phone where I enable a snoop log on the phone and then later analyze it on Wireshark. But my laptop was not able to find the log file that is allegedly saved on the phone (I also could not find it on the phone so I do not know is it an issue of finding it or whether the file was actually being created).
- Bluetooth version
  - The Ray Ban Meta is advertised to be Bluetooth 5.2 and the chipset in the glasses is Snapdragon AR1 which usually supports Bluetooth 5.0
  - But in the nRF Connect app, the glasses seem to have core 4.1, so I am unsure about the version as it could be a limitation in nRF Connect or that its configured in a way that makes it more compatible with 4.0 but supports 5.0.
  - Either way, both devices (Adafruit and Micro:bit) are able to support both BLE versions, though both support 4.0 better than 5.0 as they may not be able to sniff advanced 5.0 features. However, either way would not explain the issue I have as it would mean I should still be able to sniff something from the glasses.
- Because of the previous bulletpoint I happened to also look into the teardown of the glasses and learnt that the battery is on the right side whilst the main board is on the left side of the glasses and they made a connection going through the glasses on top of the lens (that also touches the charging area at the nose part).
- Wrote some documentation on both Adafruit and the Micro:bits as well as the errors I encountered and uploaded them.

## **What will be done next week**

- Want to check up with Nima to see how to best proceed with the issues regarding the sniffers
- Will still try to see if there are anymore online resources that may help with my situation
- Extend the documentations mainly regarding the Bluetooth sniffers but also of previous progress (the experiments).

## **Meeting**

Meeting canceled.